RAJALAKSHMI ENGINEERING COLLEGE

K S Sree Vishal 251001264-2025-IT

K2

Started on	Wednesday, 7 October 2026, 9:06 AM
State	Finished
Completed on	Wednesday, 7 October 2026, 9:07 AM
Time taken	33 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int upper_bound(int tails[], int low, int high, int target) {
    while (low < high) {
        int mid = low + (high - low) / 2;
        if (tails[mid] <= target) {
            low = mid + 1;
        } else {
            high = mid;
        }
    }
    return low;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 0;
    }

    int *arr = (int *)malloc(n * sizeof(int));
    if (arr == NULL) {
        return 1;
    }

    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int *tails = (int *)malloc(n * sizeof(int));
    if (tails == NULL) {
        free(arr);
        return 1;
    }

    int len = 0;

    for (int i = 0; i < n; i++) {
        int idx = upper_bound(tails, 0, len, arr[i]);
        tails[idx] = arr[i];
        if (idx == len) {
            len++;
        }
    }

    printf("%d\n", len);

    free(arr);
    free(tails);

    return 0;
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course